

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Summer Examination – 2023

Course: B. Tech.

Branch : FE All

Semester : II

Subject Code & Name: Engineering Mathematics-II (BTBS201)

Max Marks: 60

Date: 12-07-2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	If $\tan(A + iB) = x + iy$ then show that i) $\tan 2A = \frac{2x}{1-x^2-y^2}$ ii) $\tanh 2B = \frac{2y}{1+x^2+y^2}$	Understand (CO1)	6
B)	Show that the roots of $x^5 = 1$ are $1, \alpha, \alpha^2, \alpha^3, \alpha^4$ and hence prove that $(1 - \alpha)(1 - \alpha^2)(1 - \alpha^3)(1 - \alpha^4) = 5$	Understand (CO1)	6
C)	Prove that $\tan \left[i \log \left(\frac{a-ib}{a+ib} \right) \right] = \frac{2ab}{a^2-b^2}$	Understand (CO1)	6
Q.2	Solve Any Two of the following.		12
A)	Solve $\frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y$	Understand (CO2)	6
B)	Solve $y dx - x dy + \log x dx = 0$	Understand (CO2)	6
C)	A constant electromotive force E volts is applied to a circuit containing a constant resistance R ohm in series and a constant inductance L Henries. If the initial current is zero, show that the current builds up to half its theoretical maximum in $\left(\frac{L}{R} \log 2 \right)$ seconds.	Apply (CO2)	6
Q. 3	Solve Any Two of the following.		12
A)	Solve $\frac{d^2y}{dx^2} - 2 \frac{dy}{dx} + y = e^x + xe^x \cos x$	Understand (CO3)	6
B)	Solve $(D^2 + 2D + 1)y = e^{-x} \log x$ by method of variation of parameters	Understand (CO3)	6
C)	Solve $x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + 2y = x^2$	Understand (CO3)	6
Q.4	Solve Any Two of the following.		12
A)	Find the Fourier series of the function $f(x) = x$ in the interval $(0, 2\pi)$.	Understand (CO4)	6
B)	Find the Fourier series of $f(x) = x^2$ in the interval $-\pi < x < \pi$ and hence show that $\frac{\pi^4}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$	Understand (CO4)	6

C)	If $f(x) = \begin{cases} x & , \quad 0 < x < \frac{\pi}{2} \\ \pi - x & , \quad \frac{\pi}{2} < x < \pi \end{cases}$ then find half range Fourier sine series Hence show that $f(x) = \frac{4}{\pi} \left(\sin x + \frac{\sin 3x}{3^2} + \frac{\sin 5x}{5^2} + \dots \right)$	Understand (CO4)	6
Q. 5	Solve Any Two of the following.		12
A)	If $\vec{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and $r = \vec{r}$ then Find $\nabla \cdot \vec{F}$, where $\vec{F} = \left(\frac{x}{r}\right)\mathbf{i} + \left(\frac{y}{r}\right)\mathbf{j} + \left(\frac{z}{r}\right)\mathbf{k}$	Understand (CO5)	6
B)	Verify Green's theorem for $\oint_C (xy + y^2)dx + x^2dy$ where C is bounded by $y = x$ and $y = x^2$	Understand (CO5)	6
C)	Verify the Stokes theorem for $\vec{F} = x^2\mathbf{i} + xy\mathbf{j}$ over the square in the plane $z = 0$ bounded by the lines $x = 0, x = a, y = 0$ and $y = a$ *** End ***	Apply (CO5)	6